

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks: 30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I M. Gokul 192565067 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



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(89)

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Employee Certification

1) propositional logic.

let:

T = Kevin completed the training
 C = Kevin received certification

Rule:

$$T \rightarrow C$$

fact:

$$T$$

Goal:

$$C$$

2) CNF and Negated Goal

CNF: $T \rightarrow C$

Negated goal: $\neg T \vee C$

clauses:

$$- C$$

$$(T \vee C), T, \neg C.$$

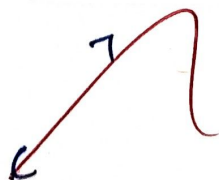
3) Resolution:

Resolve:

with:

Result:

$$(T \vee C)$$



$$C$$

Resolve:

with:

$$\neg C$$



Result:

Certification

27) Medical Attention alert

1.) propositional logic

let:

F = Ravi has a fever

M = Ravi needs medical attention

Rule:

$$F \rightarrow M$$

fact:

F

2.) CNF

$$F \rightarrow M$$

Negated goal:

$$\neg M$$

3.) Resolution:

Resolve

$$\neg F \vee M$$

with:

F

Result:

M

with:

$\neg M$

Result:



Conclusion

Ravi needs medical attention

bray late fire
propositional logic
let:
 $L = B$
 $F =$

Ravi

Library late fine

1) propositional logic

let :

L = Book was returned late

F = fine is charged.

Rule :

$$L \rightarrow F$$

fact :

L

2) CNF :

$$L \rightarrow F$$

CNF :

$$\neg L \vee F$$

Negated goal :

$$\neg F$$

3) Resolution :

Resolve :

$$\neg L \vee F$$

with :

L

Result :

F

Resolve :

F

with :

$$\neg F$$

Result :



charged

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Flight Delay Notification

Let:

D = Flight 202 is delayed
 N = Passengers are notified.

Rule:

$D \rightarrow N$

fact : D

Goal : N

2) CNF

CNF : $D \rightarrow N$

Negated goal : $\neg D \vee N$
 $\neg N$

3) Resolution:

Resolve:

with:

Result:

$\neg D \vee N$

D

N

Resolve:

N

with:

$\neg N$

Result:

$\square$

Conclusion:

passengers are notified

Employee Benefits Eligibility

1. propositional logic

let:

P = Employee finished probation

S = Employee become permanent staff

B = Employee gets full benefits

Rules:

$$P \rightarrow S$$

$$S \rightarrow B$$

fact:

P

Goal:

B

2. Convert to CNF

first rule:

$$P \rightarrow S$$

CNF:

$$\neg P \vee S$$

second rule:

$$S \rightarrow B$$

CNF:

fact:

P

3.) Negate the Goal

Goal: B

Negated goal: $\neg B$

Complete Clause:

$(\neg p \vee s)$

$(\neg s \vee B)$

p

$\neg B$

4.) Apply Resolution

Step 1:

Resolve: $\neg p \vee s$

With: p

Result: s

Step 2:

Resolve:

$\neg s \vee B$

~~Result:~~

s

B

Step 3:

Resolve:

B

With:

$\neg B$

Result:



Therefore:

The employee gets full benefits

$B = \text{True}$